

Analyzing data on the level of cognitive processes - ESCoP 2025

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Exercise 7: Signal Discrimination Model

In this exercise, we will have a look at fitting the Signal Discrimination model to a continuous reproduction task varying the set size of to be remembered color patches.

```
# empty environment
rm(list = ls())

# load libraries
library("here")
library("tidyverse")
library("brms")
library("bmm")
```

The data is part of the `bmm` package and can be loaded as `oberauer_lin_2017`.

```
sdm_data <- oberauer_lin_2017
head(sdm_data)
```

```
## # A tibble: 6 x 19
##       ID session trial set_size dev_rad col_nt1 col_nt2 col_nt3 col_nt4 col_nt5
##   <dbl> <fct>   <dbl> <fct>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1     1  1 1         1  7         0.384   0.873   0.855   2.78     2.09    1.26
## 2     1  1 2         3  3        -0.454   0.855   1.95    NA       NA      NA
## 3     1  1 3         5  5        -0.0873  0.716  -2.62   -1.03    1.38    NA
## 4     1  1 4         6  6         0.367   0.262   2.04    0.105    1.10   -0.925
## 5     1  1 5         1  1        -0.0349  NA      NA      NA       NA      NA
## 6     1  1 6         1  1         0.140   NA      NA      NA       NA      NA
## # i 9 more variables: col_nt6 <dbl>, col_nt7 <dbl>, dist_nt1 <dbl>,
## #   dist_nt2 <dbl>, dist_nt3 <dbl>, dist_nt4 <dbl>, dist_nt5 <dbl>,
## #   dist_nt6 <dbl>, dist_nt7 <dbl>
```

Your task is:

1. specify the `sdm` model object using the response error `dev_rad`.

```
sdm_model <- sdm(resp_error = "dev_rad")
```

2. estimate an `sdm` model evaluating if the parameter `c` or `kappa` changes over set size. Consider entering set size as a numeric variable assuming a linear set size effect

```
sdm_data$ss_num <- as.numeric(sdm_data$set_size)
```

```
# center on set size 1
```

```
sdm_data$ss_num <- sdm_data$ss_num - min(sdm_data$ss_num)
```

```
# sort data by predictor for faster fitting
```

```
sdm_data_sorted <- sdm_data %>% arrange(ss_num, ID)
```

```
sdm_formula <- bmf(
  c ~ 1 + ss_num + (1 + ss_num | ID),
  kappa ~ 1 + ss_num + (1 + ss_num | ID)
)
```

```
default_prior(sdm_formula, data = sdm_data, model = sdm_model)
```

##	prior	class	coef	group	resp	dpar	nlpar	lb	ub
##	lkj(1)	cor							
##	lkj(1)	cor		ID					
##	normal(0, 1)	b	ss_num			c		<NA>	<NA>
##	student_t(3, 0, 2.5)	sd				c		0	
##	student_t(3, 0, 2.5)	sd		ID		c		0	
##	student_t(3, 0, 2.5)	sd	Intercept	ID		c		0	
##	student_t(3, 0, 2.5)	sd	ss_num	ID		c		0	
##	normal(0, 1)	b	ss_num			kappa		<NA>	<NA>
##	student_t(3, 0, 2.5)	sd				kappa		0	
##	student_t(3, 0, 2.5)	sd		ID		kappa		0	
##	student_t(3, 0, 2.5)	sd	Intercept	ID		kappa		0	
##	student_t(3, 0, 2.5)	sd	ss_num	ID		kappa		0	
##	student_t(5, 2, 0.75)	Intercept				c		<NA>	<NA>
##	normal(0, 1)	b				c		<NA>	<NA>
##	student_t(5, 1.75, 0.75)	Intercept				kappa		<NA>	<NA>
##	normal(0, 1)	b				kappa		<NA>	<NA>
##	constant(0)	Intercept						<NA>	<NA>
##	tag	source							
##		default							
##		(vectorized)							
##		(vectorized)							
##		default							
##		(vectorized)							
##		(vectorized)							
##		(vectorized)							
##		(vectorized)							
##		default							
##		(vectorized)							
##		(vectorized)							
##		(vectorized)							
##		user							
##		user							
##		user							
##		user							
##		user							

3. fit the sdm to the data

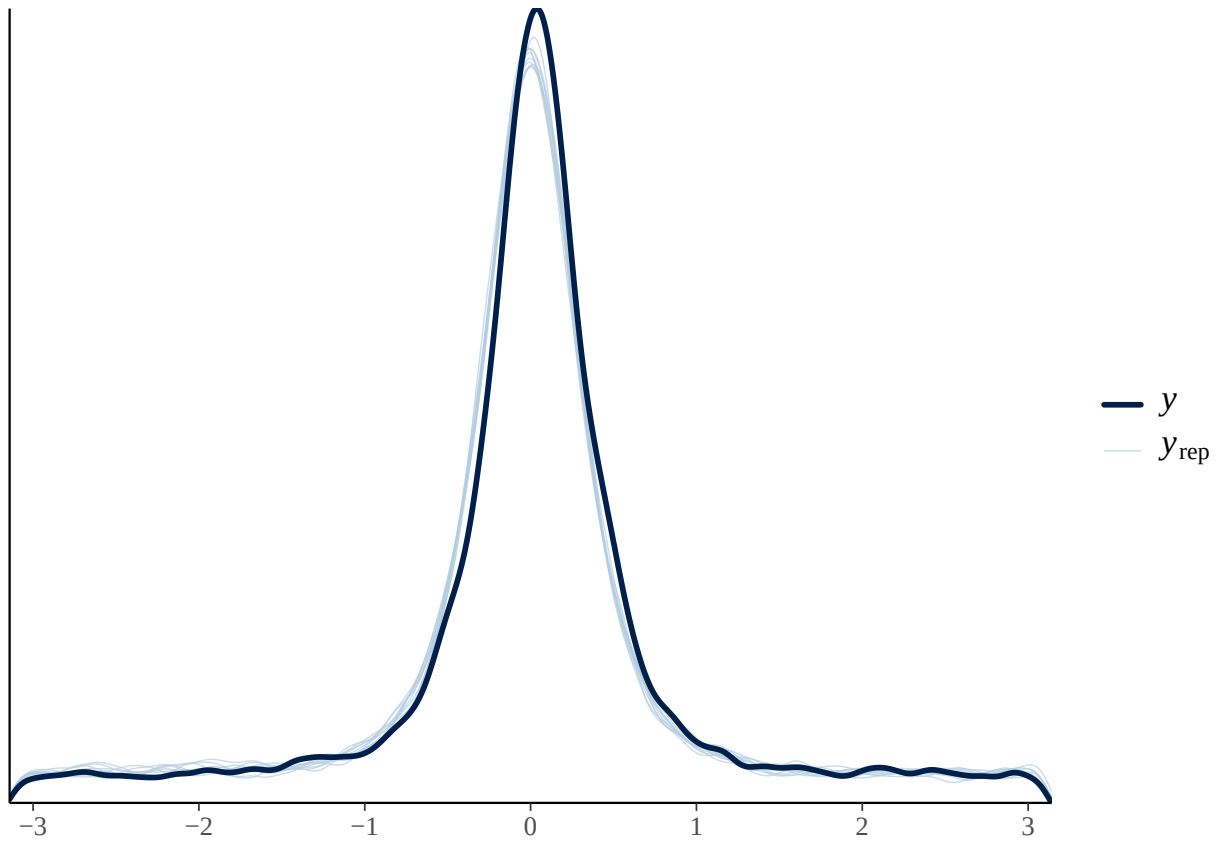
```
sdm_fit <- bmm(
  formula = sdm_formula,
  data = sdm_data_sorted,
  model = sdm_model,
  # sampler settings
  cores = 4,
  init = 0.5,
```

```

iter = 2000,
refresh = 100,
sample_prior = TRUE,
backend = "cmdstanr",
file = here("models", "exc7_sdm_setsize")
)

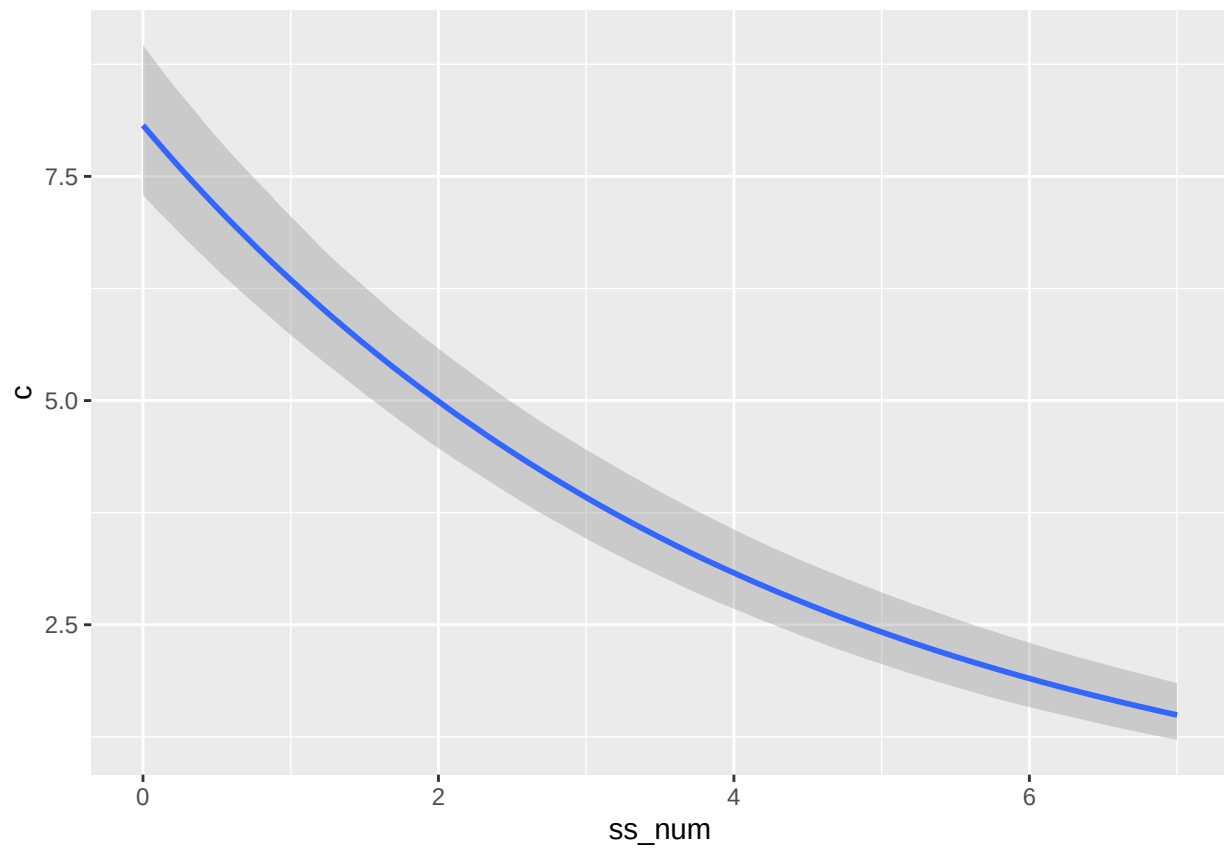
brms::pp_check(sdm_fit)

```

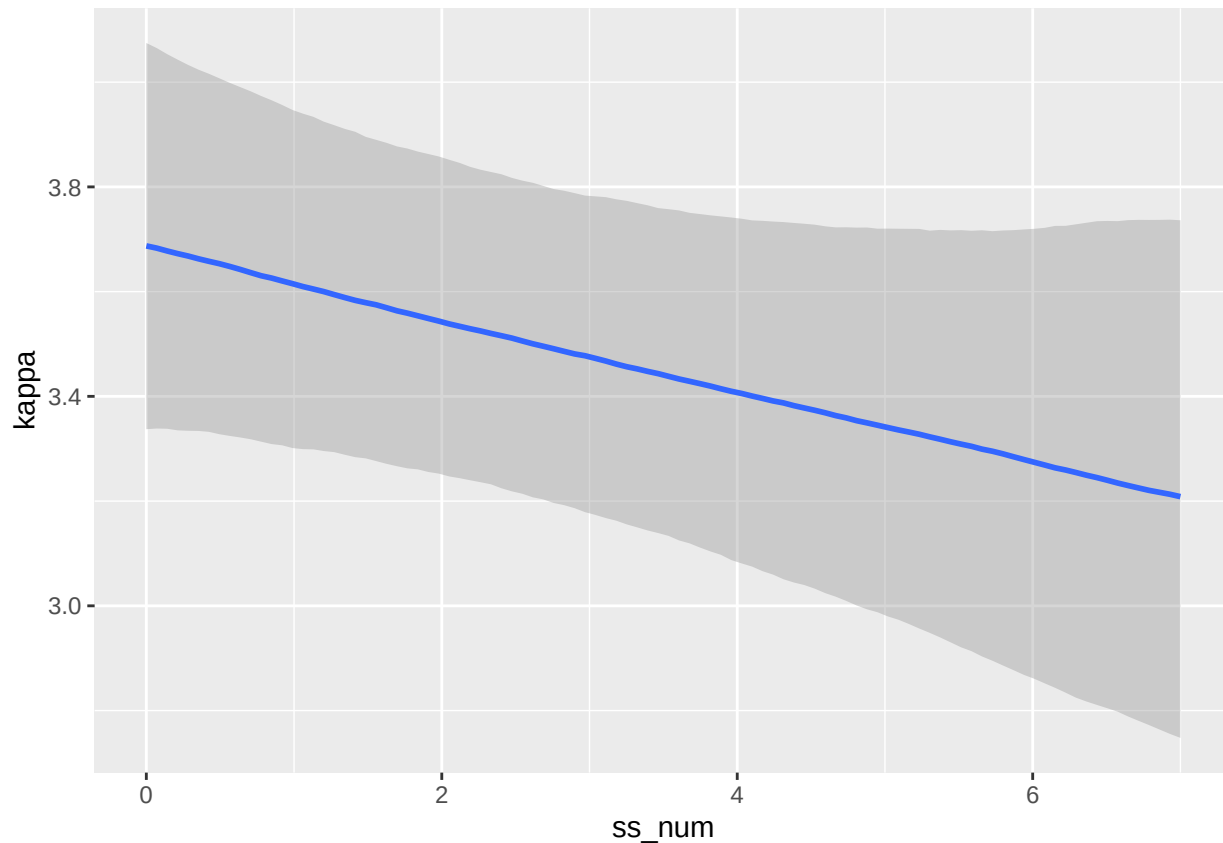


4. plot the predicted effects on c and kappa using the `conditional_effects` function

```
conditional_effects(sdm_fit, par = "c", effects = "ss_num")
```



```
conditional_effects(sdm_fit, par = "kappa", effects = "ss_num")
```

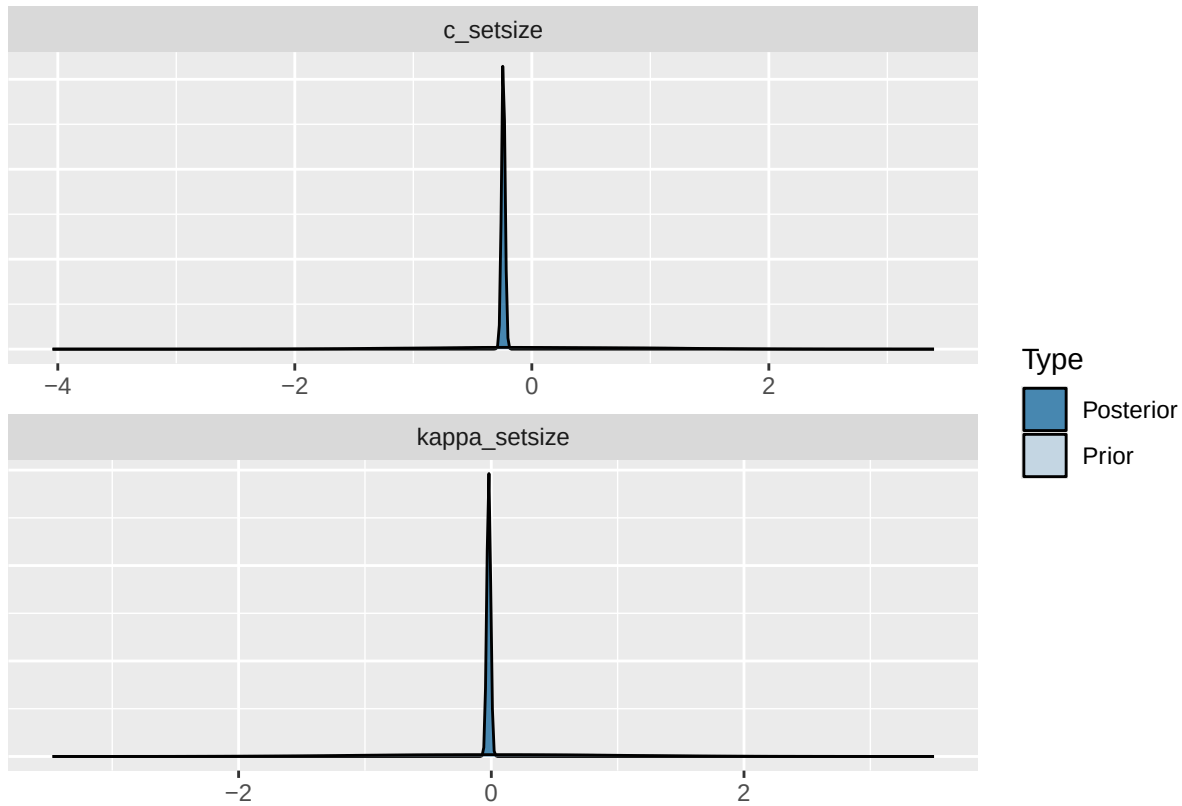


5. use the hypothesis function to test for effects of set size on c and kappa

```
ss_effects <- hypothesis(
  sdm_fit,
  c("c_setsize" = "c_ss_num = 0",
    "kappa_setsize" = "kappa_ss_num = 0")
)
ss_effects
```

```
## Hypothesis Tests for class b:
##      Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio Post.Prob Star
## 1      c_setsize   -0.24     0.01   -0.27   -0.21     0.00     0.00   *
## 2 kappa_setsize   -0.02     0.01   -0.05    0.01    29.35     0.97
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.
```

```
plot(ss_effects)
```



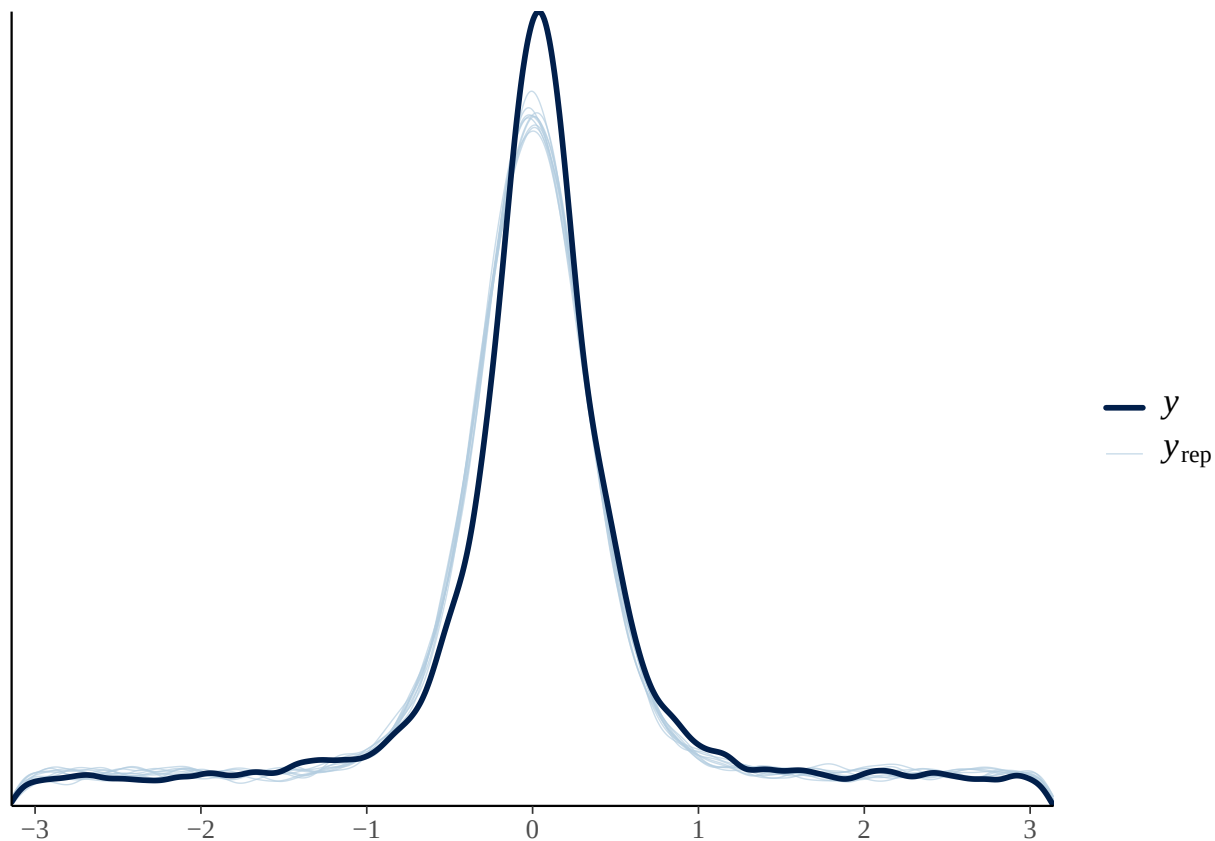
If you want compare your results to fitting a two-parameter mixture model on the data. Again, see how the proportion of responses from memory and kappa is affected by the set size variable.

```
mixture_model <- mixture2p(resp_error = "dev_rad")

mixture_formula <- bmf(
  thetat ~ 1 + ss_num + (1 + ss_num | ID),
  kappa ~ 1 + ss_num + (1 + ss_num | ID)
)

mixture_fit <- bmm(
  formula = mixture_formula,
  data = sdm_data,
  model = mixture_model,
  # sampler settings
  cores = 4,
  init = 0.5,
  iter = 2000,
  refresh = 100,
  sample_prior = TRUE,
  backend = "cmdstanr",
  file = here("models", "exc7_mixture2p_setsize")
)

pp_check(mixture_fit)
```

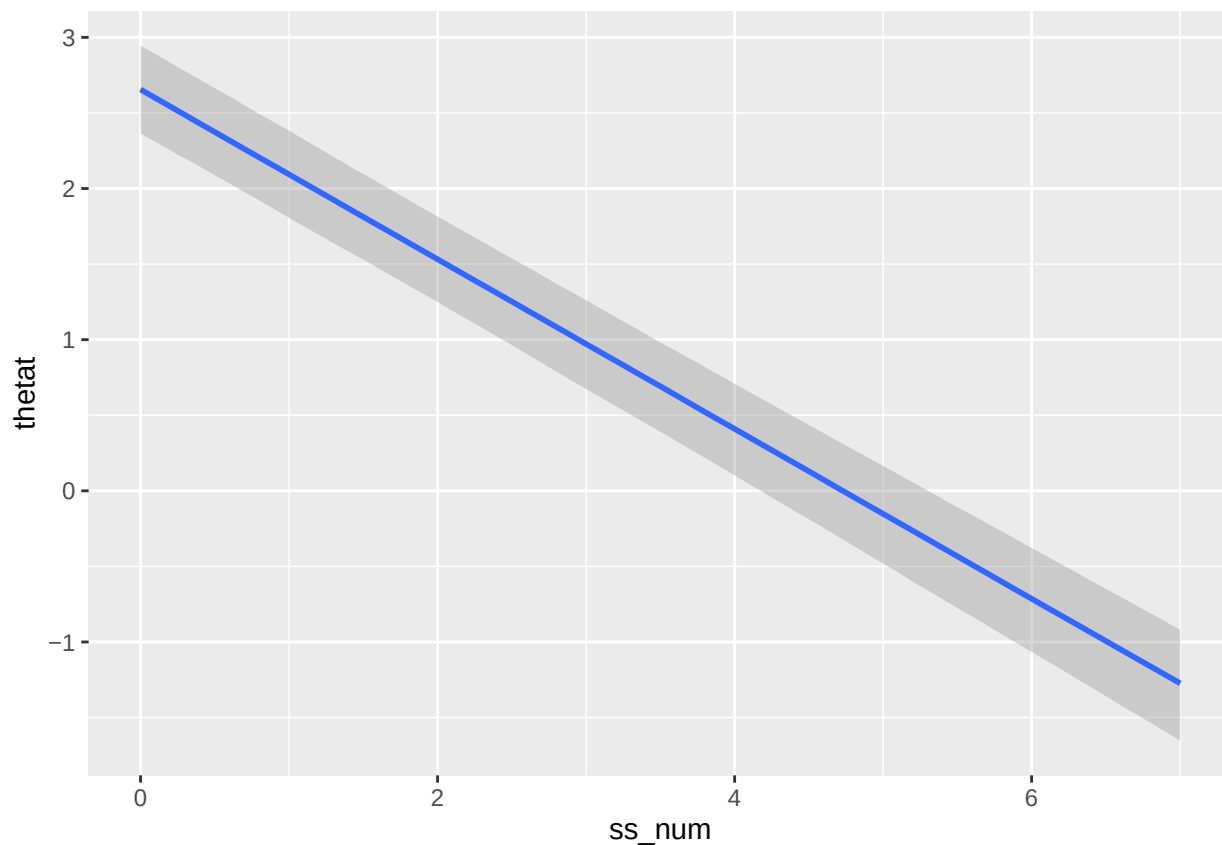


```
summary(mixture_fit)
```

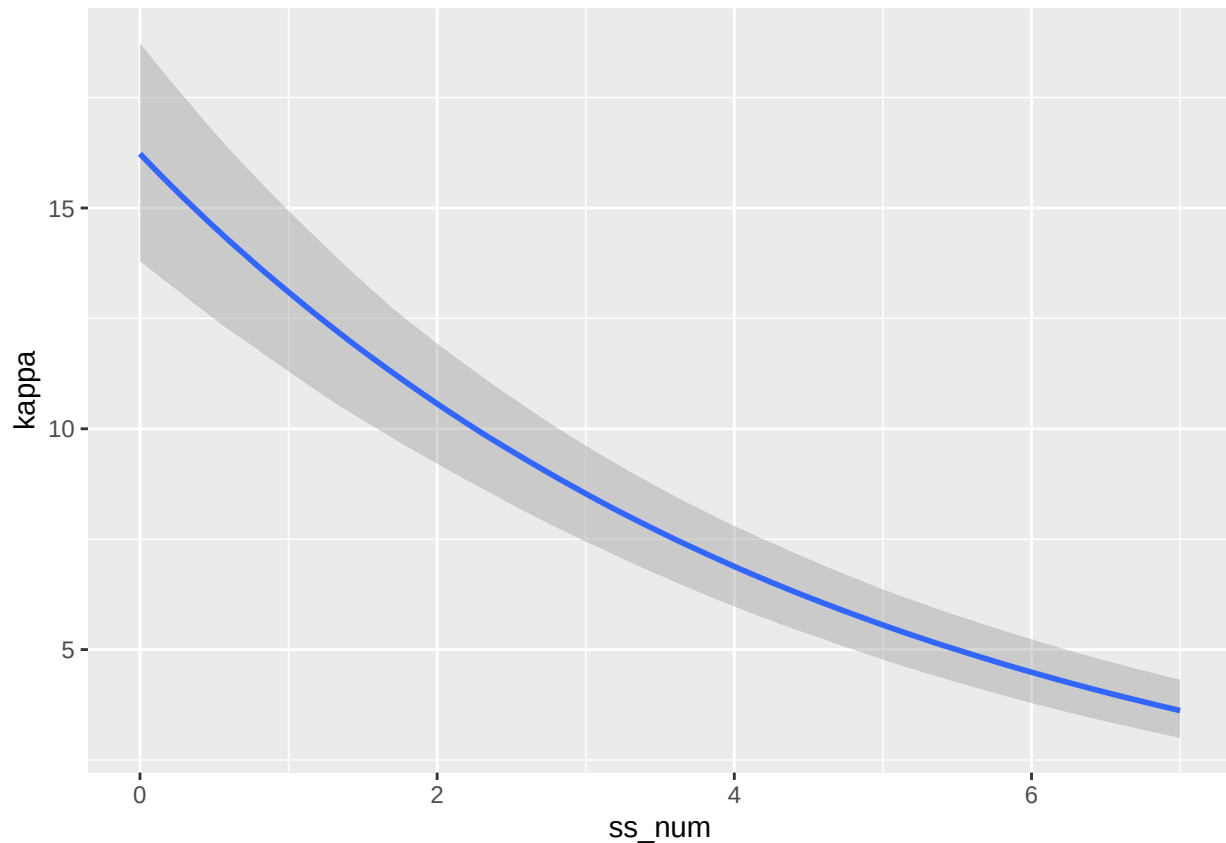
```
## Model: mixture2p(resp_error = "dev_rad")
## Links: mu1 = tan_half; kappa = log; thetat = identity
## Formula: mu1 = 0
##          kappa ~ 1 + ss_num + (1 + ss_num | ID)
##          thetat ~ 1 + ss_num + (1 + ss_num | ID)
## Data: sdm_data (Number of observations: 15200)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##        total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~ID (Number of levels: 19)
##
##          Estimate Est.Error 1-95% CI u-95% CI Rhat
## sd(kappa_Intercept)      0.30    0.06    0.20    0.45 1.00
## sd(kappa_ss_num)         0.03    0.02    0.00    0.07 1.00
## sd(thetat_Intercept)     0.56    0.13    0.35    0.85 1.00
## sd(thetat_ss_num)        0.04    0.02    0.00    0.09 1.00
## cor(kappa_Intercept,kappa_ss_num) -0.58    0.36   -0.98    0.43 1.00
## cor(thetat_Intercept,thetat_ss_num) 0.35    0.41   -0.53    0.96 1.00
##
##          Bulk_ESS Tail_ESS
## sd(kappa_Intercept)     1324    1898
## sd(kappa_ss_num)        1138    1100
## sd(thetat_Intercept)    1387    2356
## sd(thetat_ss_num)       932    1084
## cor(kappa_Intercept,kappa_ss_num) 2521    2197
## cor(thetat_Intercept,thetat_ss_num) 1970    2373
```

```
##
## Regression Coefficients:
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## kappa_Intercept      2.78     0.08   2.62   2.93 1.00     979    1534
## kappa_ss_num         -0.21     0.02  -0.24  -0.18 1.00    2432    2484
## thetat_Intercept     2.65     0.15   2.36   2.94 1.01    1014    1778
## thetat_ss_num        -0.56     0.02  -0.60  -0.52 1.00    3452    2665
##
## Constant Parameters:
##           Value
## mu1_Intercept    0.00
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
conditional_effects(mixture_fit, par = "thetat", effects = "ss_num")
```



```
conditional_effects(mixture_fit, par = "kappa", effects = "ss_num")
```

```
hypothesis(
  mixture_fit,
  c("pMem_ss" = "thetat_ss_num = 0",
    "kappa_ss" = "kappa_ss_num = 0")
)
```

Hypothesis Tests for class b:

##	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
## 1	pMem_ss	-0.56	0.02	-0.60	-0.52	NA	NA	*
## 2	kappa_ss	-0.21	0.02	-0.24	-0.18	0	0	*

'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.

'*': For one-sided hypotheses, the posterior probability exceeds 95%;

for two-sided hypotheses, the value tested against lies outside the 95%-CI.

Posterior probabilities of point hypotheses assume equal prior probabilities.